Backend

Phase 1 — API Foundation

- Audit existing FastAPI architecture
- Clean obsolete endpoints
- Define OCR API schema
- Implement `/ocr/extract`
- Implement `/ocr/extract-region`
- Implement `/ocr/export/txt`
- Implement `/ocr/correct`
- Implement `/ocr/benchmark`
- Implement API error handling
- Implement file validation
- Implement response validation

Phase 2 — Document Processing

- Image ingestion
- PDF ingestion
- Multi-page PDF processing
- Screen-capture ingestion
- Region-crop processing
- Coordinate transformation
- Normalized bounding boxes
- Page metadata
- OCR provenance metadata

Phase 3 — OCR Integration

- Connect backend to AI/ML engine
- Pass preprocessing mode
- Pass selected region
- Receive OCR blocks
- Return confidence
- Return source/model
- Return reading order
- Return normalized coordinates

Phase 4 — Export

- TXT export
- JSON export
- Copy-to-clipboard API support
- Corrected-text export

- Multi-page text export

Frontend

Phase 5 — OCR Studio

- Build OCR workspace
- Document upload
- PDF upload
- Image upload
- OCR processing state
- OCR result display
- Confidence indicators
- Printed/handwritten indicators
- Model provenance display

Phase 6 — Google Lens Interface

- Floating OCR toolbar
- One-click capture button
- Screen/window capture
- Capture preview
- Marquee selection
- Region confirmation
- Run OCR button
- OCR overlay
- Bounding-box rendering
- Coordinate scaling
- Hover OCR information
- Click-to-copy
- Copy all
- TXT download

Phase 7 — HIS Plugin

- Design plugin architecture
- Implement floating button
- Implement one-click workflow
- Implement screen capture
- Implement OCR overlay
- Ensure overlay doesn't block HIS controls
- Add minimize/maximize
- Add close button
- Add settings
- Handle screen permissions

Phase 8 — Production UI

- Loading states
- Error states
- OCR confidence UI
- Human verification UI
- Correction interface
- Document history if required
- Responsive UI
- Accessibility
- Production styling